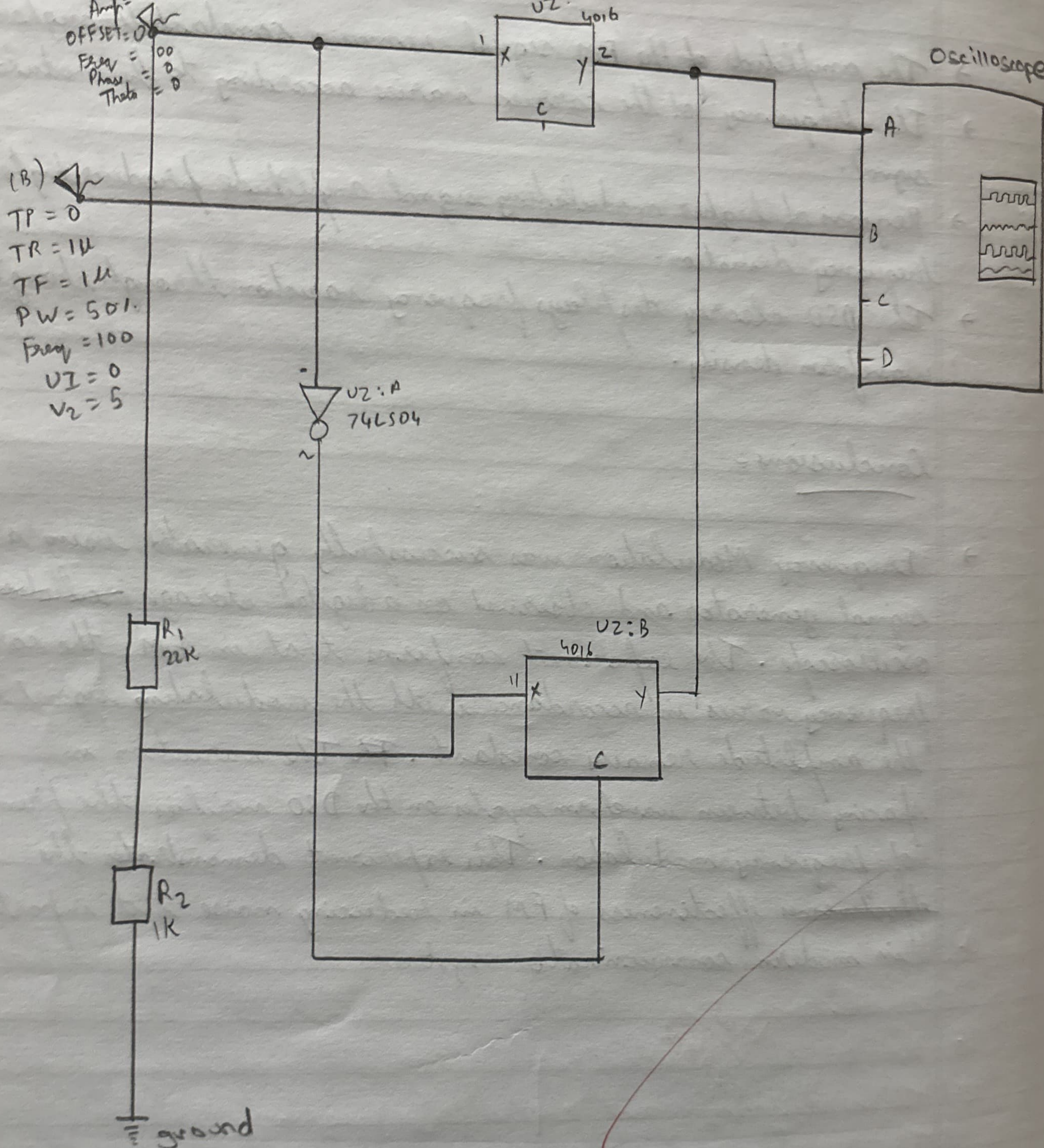


Amp
OFFSET = 0
Freq = 100
Phase = 0
Theta = 0
U2
4016
1
X
Y
2
C
Oscilloscope
A
B
C
D
(B)
TP = 0
TR = 1µ
TF = 1µ
PW = 50%
Freq = 100
V1 = 0
V2 = 5
U2:A
74LS04
R1
22K
U2:B
4016
11
X
Y
C
R2
1K
ground

Practical -3

AIM:- To study and verify the operation ~~fo~~ of a logic gate circuit using analog switches and an inverter with signal input from a function generator and analysis on a digital oscilloscope.

Objective:-

1.) To understand the working ~~th~~ of CMOS analog switches (CD4016) in digital circuits.

2.) To implement and observe signal routing / selection using analog switches controlled by logic levels.

3.) To analyse input & output waveforms using a digital oscilloscope use different control condition.

4.) To verify signal inversion using a 74LS04 hex invertor.

Theory:- CD4016 Quad ~~B~~ Bilateral switch: A chip with four electronic switches Each switch can turn ON or OFF to let a signal pass through, controlled by a digital High or Low signal 74LS04 Hex Invertor: A chip with six NOT gates. It flips a High signal to Low signal, or a Low to High.

Signal Routing :- The process of directing an electrical signal from one place to another, like choosing which path a signal should take.

Multiplexing:- Combining multiple signals into one channel.

~~E~~ Demultiplexing: Splitting one signal into multiple channels.

Oscilloscope: An instrument that shows as waveforms on a screen so you can measure voltage, frequency and timing

Triggering: A setting that ~~tha~~ tells the oscilloscope when to start drawing the waveform, so it looks stable and doesn't flicker

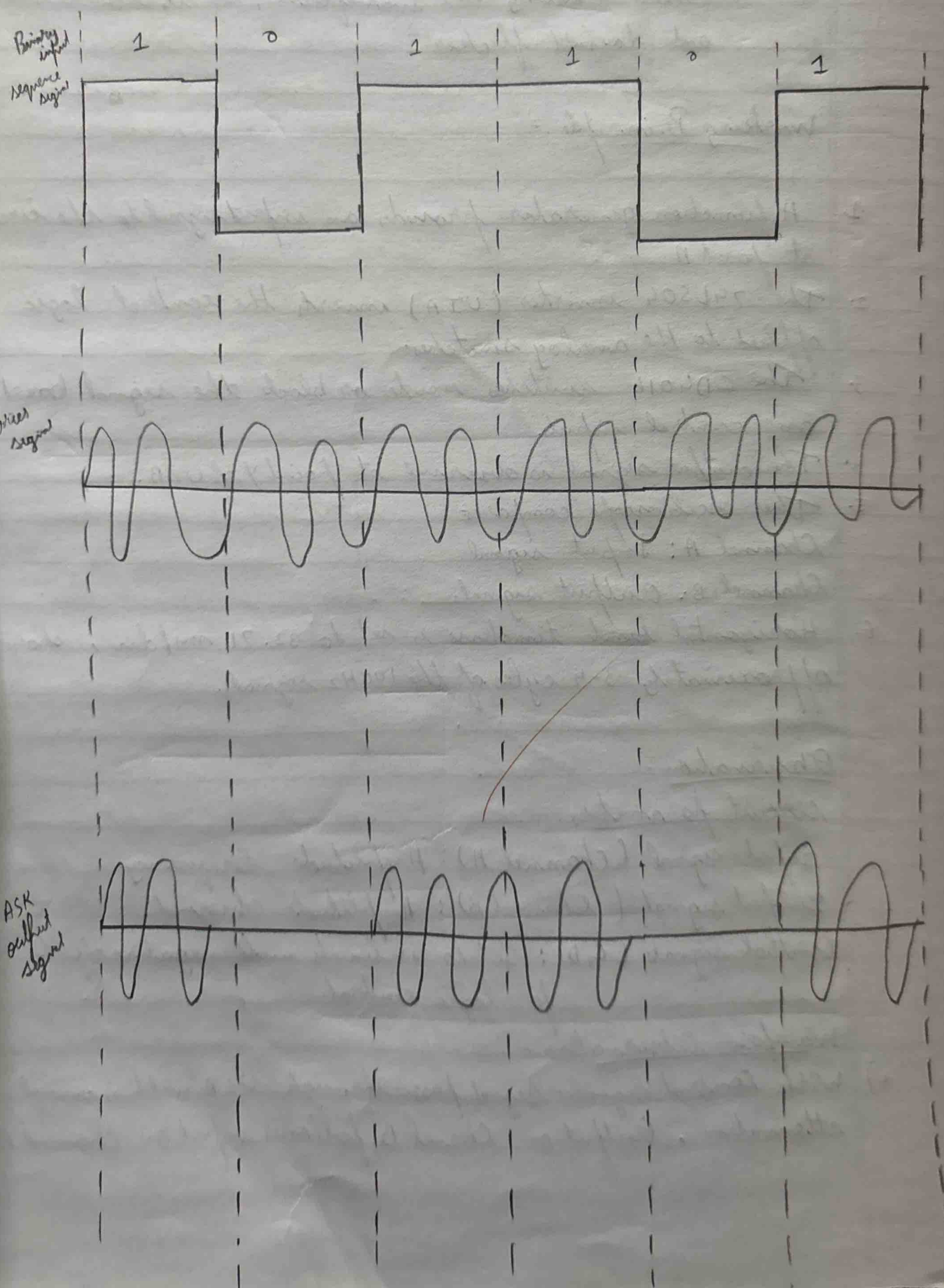
Working Principle:

1. A function generator provides an input signal to the circuit at point A.
2. The 74LS04 inverter (U1A) inverts the control logic applied to the analog switches.
3. The CD4016 switches route or block the signal based on control inputs.
4. The output signal is observed at point Y of U2B.
5. The oscilloscope compares:
   Channel A: Input signal.
   Channel B: Output signal.
6. Horizontal ~~band~~ timebase is set to 32.71 ms/div, showing approximately 3-4 cyles of the 100Hz signal.

Observation:
Circuit Parameters:
Input signal (Channel A): Amplitude, Frequency
Output signal (Channel B): Amplitude Observed.
Control signals (C, D): Set to 5v levels with negative offsets for logic control.

Waveform Observation:
1.) With Control Signal: Signal passes through U2A with minimal attenuation. Output on Chanel B follows input on channel A.

Teacher's Sign.: ____


Binary input sequence signal
1
0
1
1
0
1
carrier signal
ASK output signal

EXPERIMENT : No. Date

2. With ~~color~~ control signal : Output on channel B shows ON.
3. With inverter active : when input control to U1A is High output to switch control is Low & vice versa. This enables complementary switching between two signal paths
4. Oscilloscope readings : Time period measured, phase difference between input & output = 0° when switch is ~~on~~ ON. Signal inversion is not observed in the waveform.

Conclusion :-

→ This practical demonstrated the working of CMOS analog switches and TTL inverters for controlling signal paths. The CD4016 allowed analog switches to pass or block using digital inputs, while the 74LS04 provided complementary ~~sw~~ switching logic. Oscilloscope results confirmed correct signal transmission when ON and isolation when OFF, reinforcing concepts of signal multiplexing and digital control in mixed-signal circuits.

24/3/26

Teacher's Sign. :